
BIOLOGY

9700/42

Paper 4 A Level Structured Questions

October/November 2018

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **13** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
underline	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore
AVP	alternative valid point

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Question	Answer	Marks
1(a)(i)	<i>any two from</i> 1 deforestation or habitat, destruction / fragmentation ; 2 killed ; 3 disease ; 4 reduction in food supply ;	2
1(a)(ii)	<i>any three from</i> 1 captive breeding / AI / IVF / surrogacy ; 2 release into the wild / insurance populations ; 3 education / awareness ; 4 (work with Philippine government to) set up reserves ; 5 research, diet / habitat / breeding / behaviour / genetic diversity ; 6 veterinary care / monitor health ;	3
1(b)	<i>any three from</i> 1 hybrids formed / fewer warty pigs produced ; 2 may be sterile ; 3 genetically different (from warty pigs) ; 4 (hybrid) less adapted for natural environment ; 5 (hybrid) <i>ref. to</i> susceptibility to, diseases / parasites ;	3
1(c)	<u>Eukarya</u> ; R eukaryote(s) / Eukaryota <u>class</u> ;	2

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Question	Answer	Marks
2(a)(i)	<u>directional</u> ;	1
2(a)(ii)	<i>any three from</i> 1 lactose / milk (products), acts as a selection pressure or <i>ref. to</i> reliance on milk (products) ; 2 (selective) <u>advantage</u> to digest, lactose / milk (products) or individuals with lactose persistence have a (selective) <u>advantage</u> ; 3 (those individuals) more likely to, survive / reproduce ; or a 4 passed on the (mutated) allele (to their offspring) ; 5 over, time / many generations, the allele frequency increased ;	3
2(b)(i)	$q^2 = 58 \div 166$ or 0.349 ; A 0.35 $q = \sqrt{0.349}$ or 0.591 ; A 0.59 $p = 0.409$ or 0.41 ;	3
2(b)(ii)	<i>any two from</i> 1 test population is (too) small ; 2 test population, not representative of general population / not random / is biased ; 3 migration / ethnic origin ;	2
2(c)(i)	<i>any two from</i> 1 proteins that bind to DNA ; 2 binds to the promoter ; A enhancers 3 control, gene expression / transcription / mRNA synthesis ; 4 allow attachment of RNA polymerase to DNA ;	2
2(c)(ii)	promoter / enhancer / silencer / insulator ;	1
2(c)(iii)	1 most genes are regulatory genes (in the genome) ; or a 2 mutations in regulatory genes less likely to be, harmful / selected against / affect survival ; or a	2

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Question	Answer	Marks
3(a)(i)	57 ;	1
3(a)(ii)	different, species / genus ;	1
3(b)	parents AaHh x aahh ; gametes AH Ah aH ah ah ; offspring genotype AaHh AaHh aaHh aahh ; offspring phenotype linked to genotype white hornless white horned bezoar hornless bezoar horned ;	4
3(c)(i)	less, painful / stressful / harmful, for animal ;	1
3(c)(ii)	any one from 1 does not change other genes / only changes one gene ; 2 less time consuming / higher success rate ;	1

Question	Answer			Marks
4(a)	protein	location		4
	myosin and actin	B	;	
	actin alone	C	;	
	ATP synthase	A	;	
	ATPase	B	;	

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Question	Answer	Marks
4(b)	<i>any five from myosin binding site exposed then</i> 1 myosin head, binds to actin / forms cross bridge ; 2 ADP released causes motion of myosin head ; 3 actin moved ; 4 power stroke ; 5 ATP binds to myosin head ; 6 myosin head detaches from actin ; A cross bridge broken 7 (myosin head / ATPase) causes, hydrolysis of ATP / $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$; 8 myosin head moves back to original position / AW ; 9 (ATP needed) to pump Ca^{2+} back into sarcoplasmic reticulum ;	5

Question	Answer	Marks
5(a)	1 restriction endonuclease cuts, vector / plasmid ; A restriction enzyme 2 reverse transcriptase to make cDNA using mRNA ; 3 DNA ligase joins sugar phosphate backbone (between gene and vector) or DNA ligase forms phosphodiester bonds (between gene and vector) ;	3
5(b)	<u>marker</u> ; no fluorescence means GFP gene was deleted ; ora	2
5(c)(i)	67.6 or 68 ;	1
5(c)(ii)	higher, as expect 50% (of offspring to get GFP gene from heterozygous male) ; ecf lower, as expect 50%, (if answer to (c)(i) less than 50%)	1
5(c)(iii)	χ^2 / chi-squared ;	1

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Question	Answer	Marks
5(c)(iv)	1 10 ng mm ⁻³ ; <i>any two from</i> 2 more blastocysts ; ora 3 less toxic ; ora 4 no blastocysts seen under filter / as successful as higher concentrations / all blastocysts have deleted GFP ;	3
5(d)	<i>any three from</i> 1 lanes 1–4 show 4 kbp fragment ; 2 so technique is 100% successful ; 3 (6 kbp gene has) 2 kbp, deleted / lost ; 4 pigs (1–4) have no (normal cell surface) protein ; 5 PRRSV / virus, cannot infect the, cells / pigs (1–4) ;	3

Question	Answer	Marks								
6(a)(i)	any two from <table><tr><th>structural gene</th><th>name of gene product</th></tr><tr><td><i>lacZ</i></td><td>β-galactosidase / lactase ;</td></tr><tr><td><i>lacY</i></td><td>(lactose) permease ; A (β-galactoside) permease</td></tr><tr><td><i>lacA</i></td><td>transacetylase ;</td></tr></table>	structural gene	name of gene product	<i>lacZ</i>	β-galactosidase / lactase ;	<i>lacY</i>	(lactose) permease ; A (β-galactoside) permease	<i>lacA</i>	transacetylase ;	2
structural gene	name of gene product									
<i>lacZ</i>	β-galactosidase / lactase ;									
<i>lacY</i>	(lactose) permease ; A (β-galactoside) permease									
<i>lacA</i>	transacetylase ;									
6(a)(ii)	gene products / enzyme / protein, needed all the time ;	1								
6(a)(iii)	any two from 1 (repressor protein / gene product), binds to the operator ; 2 blocks promoter ; 3 RNA polymerase unable to bind to promoter ; 4 no, transcription / expression / activation / mRNA synthesis, of (named) structural genes ;	2								

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Question	Answer	Marks
6(b)(i)	1 only produced when, substrate / inducer / lactose, is present ; 2 (substrate / inducer / lactose) causes gene expression / gene activation / transcription / mRNA synthesis ;	2
6(b)(ii)	no waste of, amino acids / ATP / nucleotides / energy ;	1

Question	Answer	Marks
7(a)(i)	1 carbon (dioxide) fixation / carboxylation ; 2 rubisco / ribulose biphosphate carboxylase oxygenase ; 3 two named substances ;; e.g. starch, cellulose, fatty acids, lipids, sucrose 4 5 photophosphorylation ; A light-dependent stage	5
7(a)(ii)	regeneration of RuBP ;	1
7(b)	<i>any three from:</i> 1 stops <u>oxygen</u> getting to, rubisco / RuBP / bundle sheath cells ; 2 <u>oxygen</u> does not, react with rubisco / combine with RuBP ; 3 no photorespiration ; 4 no wastage of RuBP ;	3

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Question	Answer	Marks
8(a)	<i>any two from</i> 1 <i>idea that</i> DELLA proteins prevent the activation of genes (for stem elongation) ; 2 gibberellin binds to receptors (on cell surface membrane) ; 3 causes breakdown of DELLA proteins ; 4 (so) transcription / gene expression / gene activation / mRNA synthesis, can occur ; 5 AVP ; e.g. <i>ref. to</i> transcription factors / PIF	2
8(b)	<i>any four from</i> 1 seed absorbs water ; 2 embryo produces gibberellin ; 3 gibberellin, moves to / acts on / stimulates, aleurone layer ; 4 (where) production of amylase occurs ; 5 amylase, hydrolyses / breaks down, starch in endosperm ; 6 to, maltose / glucose ; 7 embryo uses sugars for, respiration / growth ; 8 AVP ; e.g. gibberellins affect, gene / synthesis of mRNA, coding for amylase	4
8(c)	<i>any four from</i> 1 both P and Q same until, day 2 / paste applied ; 2 P, greater stem length / greater height / taller, than Q (after, day 2 / paste applied) ; 3 P 35 cm and Q 15 cm / P 20 cm longer than Q, at, end (of investigation) / day 20 ; 4 P greater rate than Q / AW ; 5 comparative calculated growth rates for P and Q ; e.g. $1.75 \text{ (cm day}^{-1}\text{)}$ and $0.75 \text{ (cm day}^{-1}\text{)}$ or $1.89 \text{ (cm day}^{-1}\text{)}$ and $0.78 \text{ (cm day}^{-1}\text{)}$	4
8(d)	<i>any three from</i> 1 tall pea plants have, dominant allele / <i>Le</i> ; 2 (which codes for) enzyme that produces active gibberellin ; 3 dwarf pea plants (only) have, recessive alleles / <i>le</i> ; 4 (so) no (active) gibberellin formed ; 5 GA_1 is the active form of gibberellin ;	3

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Question	Answer	Marks
9(a)	<i>any six from</i> 1 different substrates have different numbers of, hydrogens / C-H bonds ; 2 lipids have (relatively) more, hydrogens / C-H bonds (than carbohydrates or proteins) ; 3 hydrogens / C-H bonds, located in fatty acid (tails of lipids) ; 4 breakdown / oxidation, of substrate provides hydrogen (atoms) ; 5 for reduction of, NAD / FAD ; 6 (reduced, NAD / FAD) provides / releases, hydrogen to ETC ; 7 hydrogen (dissociates) into protons and electrons ; 8 <i>ref.</i> energy used to set up proton gradient ; 9 chemiosmosis / oxidative phosphorylation / AW ; 10 (so) more, ATP / energy, from lipids per unit mass (than, carbohydrates / proteins) or lipids, more energy dense / have higher (relative) energy value ;	6
9(b)	<i>RQ</i> 1 (ratio of) carbon dioxide given out divided by oxygen taken in ; 2 <i>ref.</i> volume / moles ; R amount 3 per unit time ; <i>any eight from investigation</i> 4 use respirometer ; 5 seeds placed on, mesh / gauze ; 6 KOH / NaOH / sodalime, to absorb carbon dioxide ; 7 manometer / capillary tube / syringe ; 8 movement of fluid (in manometer / capillary tube / syringe) = uptake of oxygen ; 9 keep, temperature / air pressure, constant ; 10 measure oxygen uptake after certain time ; 11 repeat without KOH / NaOH / sodalime ; 12 difference in manometer readings due to carbon dioxide given out ;	9

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Question	Answer	Marks
10(a)	<i>any eight from</i> 1 sense organ / receptor, detects stimulus ; 2 e.g. light, sound, heat ; 3 <i>idea of</i> very strong stimulus ; 4 action potential generated in sensory neurone ; 5 sensory neurone connects to spinal cord ; 6 synapse with, relay / intermediate, neurone or action potential passes to, relay / intermediate, neurone ; 7 (relay / intermediate / sensory, neurone) synapse with motor neurone or action potential passes to motor neurone ; 8 effector / muscle ; 9 response / described ; <i>plus</i> 10 fast(er) ; 11 automatic / involuntary / AW ; 12 response always the same / stereotypic ; 13 protects from harm ;	9
10(b)	<i>any six from</i> 1 (sheath) insulates axon / stops passage of ions ; 2 gaps / nodes of Ranvier ; 3 1–3 mm intervals ; 4 passage of ions can occur (at nodes) ; 5 depolarisation / action potentials, only occur at nodes ; 6 local circuits (between nodes) ; 7 saltatory conduction ; A description 8 faster (speed of nerve impulse) ; 9 AVP ; e.g. detail of structure of sheath	6